Eliminating z from (2) and (3)

$$\begin{array}{lrl} (2)\times 3 \Rightarrow & 21x - 6z &= 126 \\ (3)\times 2 \Rightarrow & 4y + 6z &= 156 \quad (+) \\ \hline & 21x + 4y &= 282 \\ (1)\times 4 \Rightarrow & 12x - 4y &= 48 \quad (+) \\ \hline & 33x &= 330 \quad \text{so, } x = 10 \end{array}$$

Substituting $x = 10$ in (1), $30 - y = 12$ we get, $y = 18$

Substituting $x = 10$ in (2), $70 - 2z = 42$ then, $z = 14$

$\therefore\ x = 10,\ y = 18,\ z = 14$.

Example 3.8 Solve : $\frac{1}{2x} + \frac{1}{4y} - \frac{1}{3z} = \frac{1}{4}$; $\frac{1}{x} = \frac{1}{3y}$; $\frac{1}{x} - \frac{1}{5y} + \frac{4}{z} = 2\frac{2}{15}$

Solution Let $\frac{1}{x} = p$, $\frac{1}{y} = q$, $\frac{1}{z} = r$

The given equations are written as

$$\begin{aligned} \frac{p}{2} + \frac{q}{4} - \frac{r}{3} &= \frac{1}{4} \\ p &= \frac{q}{3} \\ p - \frac{q}{5} + 4r &= 2\frac{2}{15} = \frac{32}{15} \end{aligned}$$

By simplifying we get,

$$\begin{aligned} 6p + 3q - 4r &= 3 && \ldots(1) \\ 3p &= q && \ldots(2) \\ 15p - 3q + 60r &= 32 && \ldots(3) \end{aligned}$$

Substituting (2) in (1) and (3) we get,

$$\begin{aligned} 15p - 4r &= 3 && \ldots(4) \\ 6p + 60r &= 32 \text{ reduces to } 3p + 30r = 16 && \ldots(5) \end{aligned}$$

Solving (4) and (5),

$$\begin{array}{rl} 15p - 4r &= 3 \\ 15p + 150r &= 80 \quad (-) \\ \hline -154r &= -77 \end{array}$$

we get, $r = \frac{1}{2}$

Substituting $r = \frac{1}{2}$ in (4) we get, $15p - 2 = 3 \Rightarrow p = \frac{1}{3}$

From (2), $q = 3p$ we get $q = 1$

Therefore, $x = \frac{1}{p} = 3$, $y = \frac{1}{q} = 1$, $z = \frac{1}{r} = 2$. i.e., $x = 3, y = 1, z = 2$.

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